

Facility location allocation modelling for bio-energy system in Anambra State of Nigeria: Integration of GIS and location model

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ABSTRACT

The adoption of appropriate waste management strategy has been considered as a lasting solution to part of the environmental problems facing Anambra State of Nigeria. Implementation of spatial information technologies such as Remote Sensing (RS) and Geographic Information System (GIS) in addressing the issue of waste management has been identified as an appropriate technology. The limitations of the use of GIS alone or only location mathematical models in effective suitability analysis has been observed by many researchers, therefore this study adopted the integration of geospatial technology with Set Cover Location Model (SCLM) and facility location allocation modelling as a holistic approach to the problem. The result of the study shows that a minimum of three biogas processing plants is needed to maximally process all the wastes in the study area. The output of the allocation and location model shows that a processing plant with about 5.6 Tons, 1.7 Tons and 57 Tons capacity is best located at Ajali, Ihiala and Abagana towns respectively in the study area. It is recommended that strategic waste management approach as used in this study will be a veritable tool for energy solution in the study area.

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1. Introduction

In the present day, the need and practice of reclamation of bio-wastes produced from animal waste is gaining approval from developed and developing countries of the world as a result of the impact of climate change on the human environment. Small scale biogas plants are used on farm level across the country by few farmers. The current trend towards sustainable renewable energy in developing countries calls for a systematic approach of farm waste management and treatment, which can only be achieved by strategic planning. Few of the key parameters influencing the viability of community biogas digesters in Anambra State will be the distances between bio-waste sources and central digesters; feed in tariffs; willingness to invest in biogas production industry; and maintenance cost.

Due to high geographical dependence of biomass feed stocks, implementation of spatial information technologies such as RS and GIS in addressing the issue of waste management appears to be an appropriate methodology [1]. With the growing interest in exploring renewable energy, GIS has proved to be an effective tool to address issues related to biomass availability and biomass

logistics [2]. There have been moves towards a closer integration of GIS data models with transportation planning and modelling. In many cases this involved a re-purposing or re-focussing of tools and models which already existed within GIS [3]. The association of spatial optimization models with GIS can help to formulate and develop planning options which try to maximize or minimize some objective functions.

Several studies have been done using GIS in solving environmental spatial related problems. Haddad and Anderson [4] used GIS technology to identify potential locations in a Mid-western region for collection and storage of corn stover for use as biomass feedstock. Thomas et al. [5] presented an analysis of the spatial supply and demand relationships for biomass energy potential for England, using GIS mapping techniques. Uyan [6] studied the determination of suitable site for solar farms using GIS and Analytical Hierarchy Process (AHP) in their study area. The final index model was grouped into four categories as “low suitable”, “moderate”, “suitable” and “best suitable” with an equal interval classification method in Karapinar region, Konya/Turkey. While Höhn et al. [7] analyzed the spatial distribution and amount of potential biomass feedstock for bio-methane production and optimal locations, sizes and number of biogas plants in Southern Finland in the area of three regional waste management companies. Thus GIS is a veritable tool in solving

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geospatial related problems of waste management.

Many approaches have been employed in site suitability analysis and location modelling. Most of the approaches used in location modelling heavily rely on mathematics and optimization techniques. Some researchers have determined their best locations as pure mathematical problem [8,9]. Xie [10] agreeably asserted that the optimal locations for bio-refineries depend on a number of other issues that are difficult to quantify and model mathematically, that site selection is a spatial problem, mathematics and optimization techniques are often inadequate to offer acceptable solution because of their failure to incorporate all relevant aspects of the problem in the overall framework. Thus location problems are best solved using spatially added tools and programs with mathematical modelling.

Another important aspect of site suitability analysis is location-allocation modelling which seeks to match supply with demand. It seeks to identify the set of N optimal locations by considering the spatial distribution of the potential locations (supply) in relation to spatially distributed demand. Having realized that resources (facilities) are insufficient to achieve total coverage within bio-energy distance goal, the decision would be to cover as many as possible demand points using limited resource/facilities. That is, facing the reality of an insufficient number of bio-energy facilities, one may relinquish his goal of total coverage within a demand region and attempt instead to locate the facilities in such a way that only few supply points (waste generating center) as much as possible lie outside the desired service distance. This problem is designated as SCLM in operations research. SCLM maximizes the amount of covered demand/supply within the acceptable service distance by locating a given fixed number of facilities.

It is extremely difficult economic-wise to have a full coverage in facility location modelling in the study area due to dispersive state of agricultural waste generation points. Hence, SCLM was used in this study to determine minimum suitable locations that will enhance maximum coverage. The objective of this research work is to use innovative approach in solving location suitability problem by integrating geospatial technology, SCLM and location-allocation model for optimized locations considering spatial, socio-economic and technological advantages.

2. Materials and methods

2.1. The study area

Anambra State is the oldest state in South-Eastern Nigeria. The Capital and the Seat of Government is Awka. Onitsha and Nnewi are the biggest commercial and industrial cities, respectively. Boundaries are formed by Delta State to the west, Imo State and Rivers State to the South, Enugu State to the East and Kogi State to the North. The State is divided into 21 Local Government Areas (LGA) with Awka as its state capital. The study area is located between Latitudes $05^{\circ}42'56''N$ and $06^{\circ}45'34''N$ and Longitude $06^{\circ}37'30''E$ and $07^{\circ}25'30''E$. Fig. 1 below shows the map of the study area.

2.2. Data management software

Lat Long converter software was used in computation of origin destination cost matrix used in the location and allocation modelling. Microsoft Excel Solver was used in solving the SCLM and location/allocation model. ArcGIS 10 software was used to perform all the GIS operations. It was also used to analyze all the factors represented by GIS thematic layers such as the buffering of waste generation points and to produce the land suitability map.

2.3. GIS suitability analysis

2.3.1. Transportation assessment

There is need to develop a method capable of integrating biomass availability; energy demand; and biogas production in a realistic dynamic geographical model: such that conclusions can be drawn on mainly the sustainability; additionally on the efficiency and economy of biogas production in the study area. A fundamental difference between GIS-based and non-GIS models is that datasets in GIS models such as geographic distribution of biomass and suitable site graphic model for the location in the study area can be layered and geographically integrated.

The multi-layer datasets allow spatial manipulation such as extraction of biomass availability information within different driving distances from biogas plant locations. The importance of transportation costs, and the nature of commodities transported usually have major influence on the economy of bio-energy siting. The waste generation centers were buffered to ensure that transportation cost is minimized in the cause of supplying the bio-energy plants with agricultural wastes; batch buffer of 5 km, 10 km, 15 km and 20 km was used as driving distance to processing plants.

2.3.2. Land use suitability assessment

The land use suitability map was obtained by excluding unwanted areas identified as constrain map. These are areas prone to flooding, areas that are close to residential areas, water bodies, etc. The Slope layer, Hillview layer and Elevation layers were overlay at 30%, 40% and 40% weight respectively using weighted Overlay tool in ArcGIS. Some sites were selected as preliminary potential location for bio-energy processing plant; the site selection was based on suitability analytical studies; using hotspot analysis and land suitability analysis. This is however beyond the scope of this work.

2.4. Set Cover Location Model Problem Formulation

2.4.1. Problem description

Based on the available statistics on agricultural waste generation points in the state, there exist many livestock waste generating centers in each LGA, except in two LGAs. There is need to estimate best sites for allocation/transportation of these wastes, these locations are referred to as processing sites, while the waste generating sites are referred to as supply points. Assuming that a given processing site can process a given amount of waste from a radius of approximate 10 km; this represent the capacity of bio-energy plant opening in a given location. Each bio-energy processing center can accommodate and process bio-wastes with facilities at fixed cost according to a scale. The goal here is to determine which bio-energy center to open as well as the best allocation of wastes sources to the opened bio-energy processing centers.

2.4.2. Problem Formulation for maximal Set Cover Location Problem

The SCLM was applied to Anambra agricultural waste generating centers, the map of the study area showing the various agricultural waste (poultry and paunch wastes) generating centers in the State is as shown in Fig. 2. The dispersive features of the waste generating centers can be visualized from the Figure; the failure to concentrate farming activities (and hence waste generation) in a region as seen in the Figure could be peculiar problem associated with poor planning in developing countries.

Fig. 2 shows the distribution of waste generation sites in the study area, it could be seen that there is concentration of waste generation sites at the central region of the study area, while the upper North of the Figure shows high dispersive waste generation points. It is worthy to note that the North/West region is

characterized by absence of agricultural waste generation point; this could be as a result of swampy nature of the area. The dispersive otherwise clustering of the agricultural generation sites necessitates the integration of geospatial technology with location model as already stated. Based on GIS operation's results and preliminary studies conducted in the study area, suitable location sites were selected. The site selection was based on geospatial suitability analysis studies from the previous work of

Chukwuma et al. [11].

Site collection radius of 10 km was used as the maximum distance of transportation for economic reasons. The model was formulated by considering a total of ten (10) collection radii where because of overlapping of the various collection radius, minimum potential sites is to be selected in such a way that will minimize distance and will enhance maximal covering of waste generation centers. Since each radius of collection has its own potential

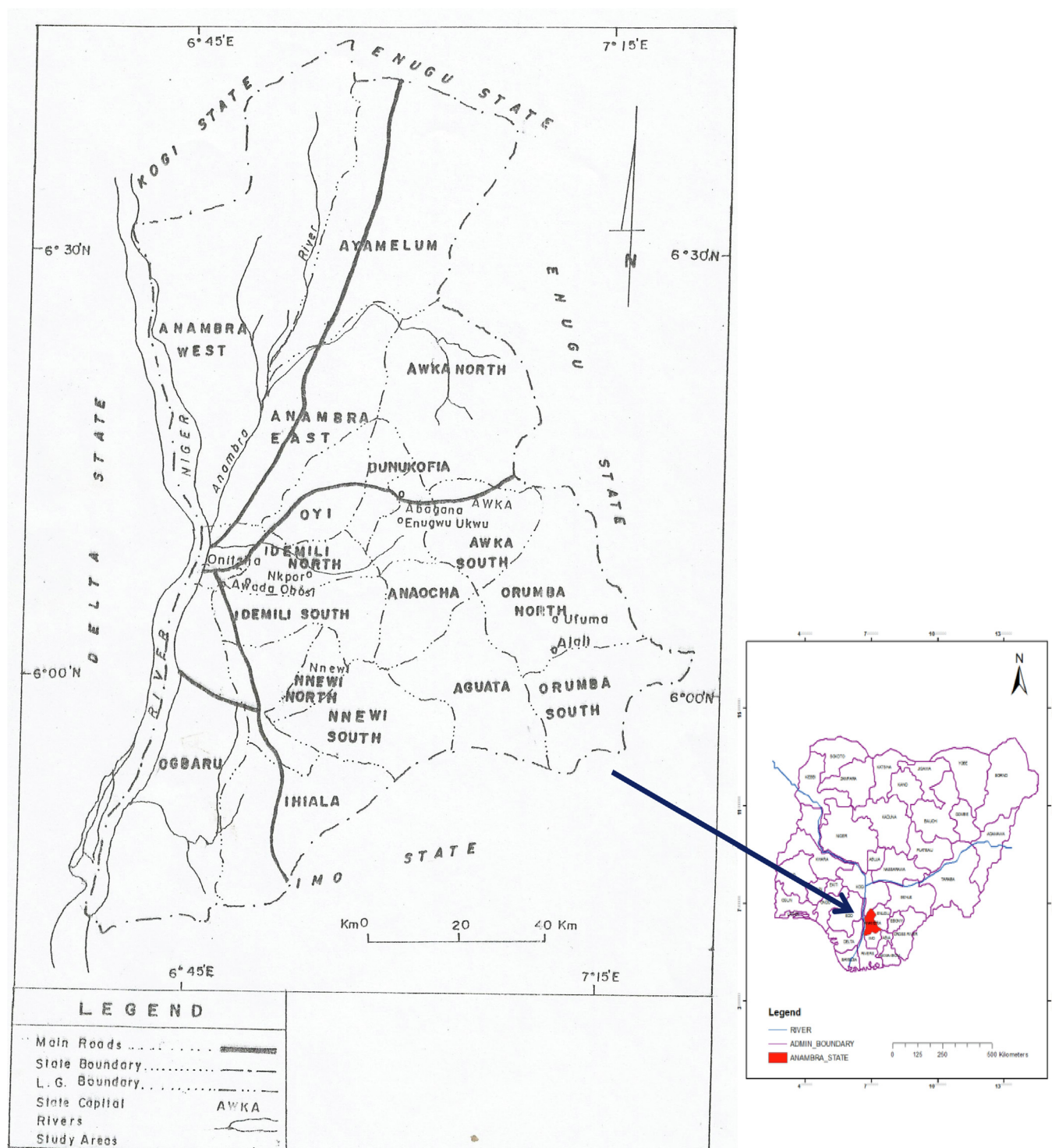


Fig. 1. Map of Anambra state of Nigeria.

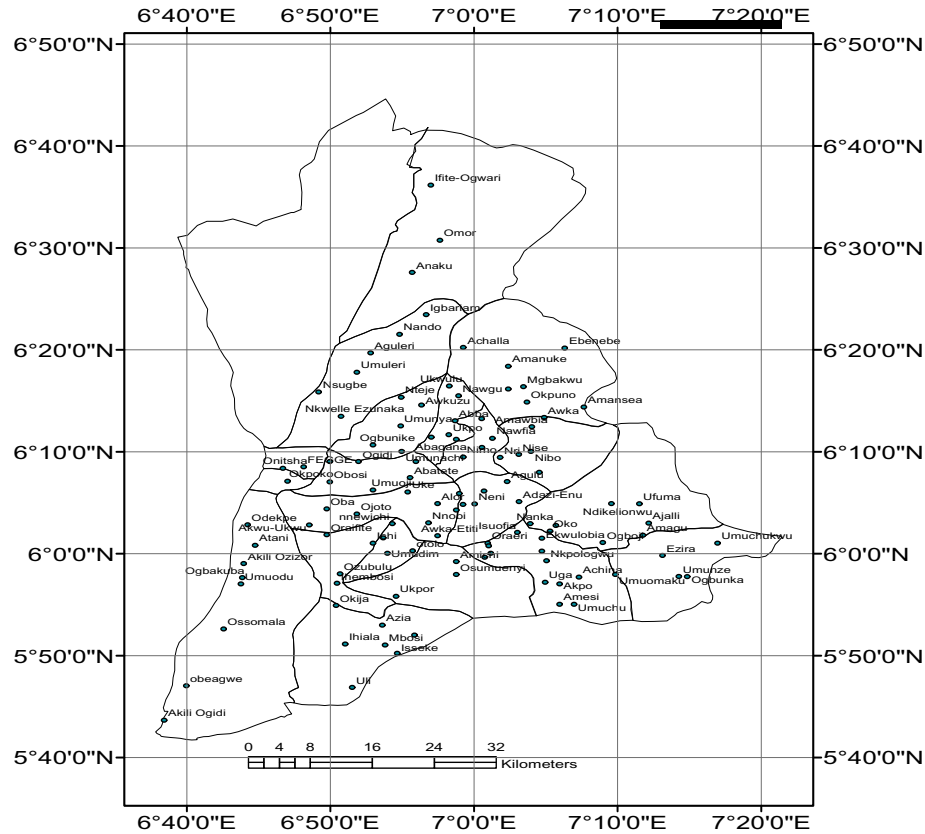


Fig. 2. Agricultural waste generation centers in the Study area.

biomass quantity, the goal of SCLM is to locate the minimum number of potential processing centers from the overlapping radii. In general, two radii connecting each other constitutes an interception set, if a waste generation point can transport its waste from two overlapping radii say 1 to 2, then distance is minimized; if otherwise distance is not minimized. By adopting set concept, the intercept/overlap of any waste collection radial to others is such that opening bio-energy processing center say in radius 1 can satisfy itself and any given number of intercepts at minimized distance. The locations of the proposed waste processing site (collection radius) are Onitsha (1), Orafite (2), Awka (3), Ihiala (4), Ajali (5), Akpo (6), Abagana (7), Adazi-ani (8), Aguleri (9) and Omor (10). The selection was done in such a way as to cover the entire waste generating centers. The Tenth collection radius was however removed considering the fact that there was no overlapping of the collection radial with others which indicates high cost of transportation of its biomass to processing centers and the condition of the SCLM that requires overlap. Fig. 3 below shows the waste collection radial for proposed suitable locations for the bio-energy plant.

Considering collection radius overlap of the potential waste processing sites as shown in the Figure. The subsets (potential sites), the covers (collection radius) and the quantity of wastes within the collection radius were determined as shown in Table 1.

The SCLM was used to determine the minimum number of processing sites needed to maximally process all the wastes in the study area. The aim of the mathematical formulation is to find the minimum number of waste processing centers that will cover majority of the waste generating centers for 9 subset indexed on $i = 1, 2, 3 \dots 9$.

Decision Variable

$X = 1$, if subset i is selected and 0 otherwise. $i = 1, 2, 3 \dots n$. For this application: $x = 1$, if a bio-energy processing potential site is opened/selected and 0 otherwise.

Parameter

$A_i = 1$, if bioenergy center opened in location i can cover location j , and 0 otherwise

Constraint

Every bio-waste source in proximity to selected processing sites must be served (or covered) by at least one bio-energy processing center.

Objective Function

$$\text{Min. } 12.74x_1 + 6.09x_2 + 10.59x_3 + \dots 3.4x_9 \quad (1)$$

Subject to

$$x_1 + x_2 + x_7 \geq 1 \text{ (Onitsha)}$$

$$x_1 + x_2 + x_4 \geq 1 \text{ (Orafite)}$$

$$x_3 + x_7 + x_8 \geq 1 \text{ (Awka)}$$

$$x_7 + x_9 \geq 1 \text{ (Aguleri)}$$

$$x_i \in (1.0) \forall i$$

The total biomass in tonnes in each subset was used as cost function in the formulation of the objective function.

2.5. Location allocation modelling

The location allocation model was further used to determine the allocation of the various wastes to the three multiple facility locations obtained using the SCLM as noted above. Further work in this

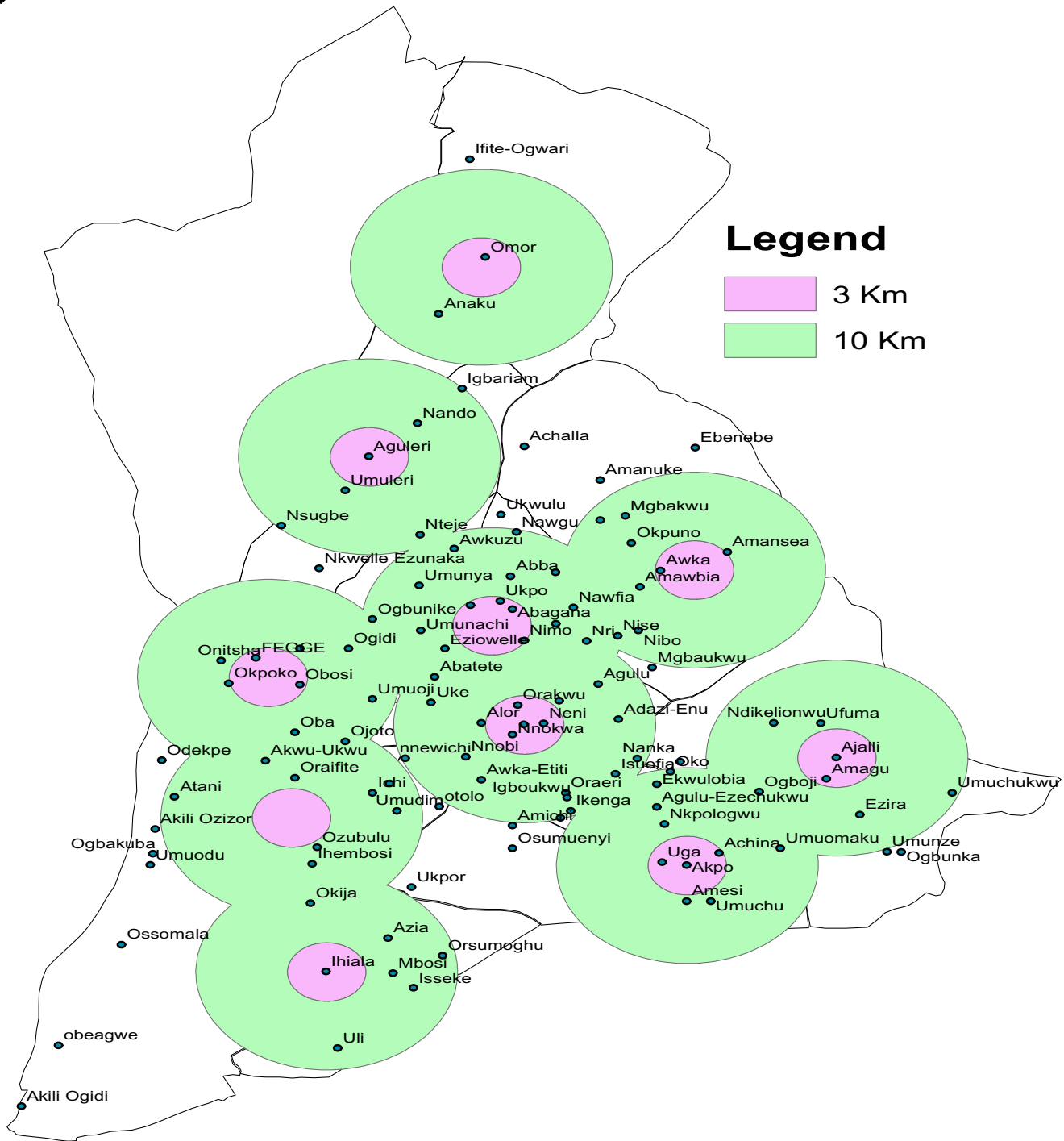


Fig. 3. Waste collection radial for proposed suitable locations.

study is to ensure that the various waste generating sites is allocated to an opened waste processing facility considering economic advantage; hence the use of location allocation problem model known as the Fixed Charged Facility Location Model (FCFLM), this is shown in equation (2):

Objective Function

$$\text{Min. } \sum_{i=1}^n f_i y_i + \sum_{i=1}^n \sum_{j=1}^m C_{ij} X_{ij} \quad (2)$$

where f_i is annualized fixed cost of biogas plant.

C_{ij} = Cost incurred from supply point to the processing point.

$y_i = 1$ if the plant i is open, 0 otherwise.

X_{ij} = Quantity shipped from demand point i to processing plant j .

The variant of FCFLP suited for bio-energy plant location optimization in this study is given as:

$$\text{Min. } \sum_{i=1}^n f_i y_i + \sum_{i=1}^n \sum_{j=1}^m d_{ij} P_{ij} \quad (3)$$

Table 1

Subset and covers used for the set covering location model and Waste generation capacity of all radii.

S/ N	Subset	Covers	Quantity of waste within radius of collection (Tons) P_{ij}
1	1 (Onitsha)	1,2,7	12749.4
2	2 (Orafite)	1,2,4	6086.2
3	3 (Awka)	3,7,8	10592.004
4	4 (Ihiala)	2,4	1752.2
5	5 (Ajali)	5,6	1729.4
6	6 (Akpo)	5,6,8	3851.3
7	7 (Abagana)	1,3,7,8,9	12976.1
8	8 (Adazi-ani)	6,7,8	11063.5
9	9 (Aguleri)	7,9	3429.1

d_{ij} represent the distance from bio-waste source i to the chosen processing plant j , this could be seen as shipping/transportation cost function.

Where P_{ij} represent the quantity of bio-waste conveyed from supply point to processing site. We have two sets here, the first is the processing set radius and the second is the supply set radius.

X is the processing set radius indexed on $i = 1$ to 3. C is the sets of supply radius whose wastes must be processed indexed on $j = 1$ to 9.

C_j is the waste generation capacity of radius j . K_i is the capacity of the processing waste centred that is to be opened.

y_i serves as decision variable for opened and closed set, it is 1 if opened and 0 otherwise, the value is predetermined in this study.

Subject to:

$$\sum_{i=1}^n x_{ij} = C_j \text{ for } j = 1 \dots m$$

This is the demand constraint

$$\sum_{j=1}^m x_{ij} \leq k_i y_i \text{ for } i = 1 \dots n$$

This is the capacity or supply constraint

$$y_i \in \{0, 1\} \text{ for } i = 1 \dots n; x_{ij} \geq 0$$

To approximate the annualized fixed investment cost of the biogas plant, f_i was determined using several economic estimation mathematical models which is beyond the scope of the work, further details can however be found in Chukwuma and Orakwe [11]. The origin cost destination matrix was developed for all the sets in linear distance of Kilometres and used in the location allocation modelling. The origin cost destination matrix of all the 10 locations is shown in Table 2:

The transportation cost function (d_{ij}) of selected locations used

in the location/allocation modelling is highlighted in the Table 2 in bold and italics; this was used in equation (3) above. The goal to determine the allocation of collection set radius to processing set radius for the system is shown in equation (4) as stated in equation (3) above. The specific objective function for the problem is:

$$\begin{aligned} \text{Min } & 4000 \times y_1 + 380 \times y_2 + 120 \times y_3 + P_{11} \\ & + 13.23P_{12} 34.92P_{13} + \dots + 19.08P_{310} \end{aligned} \quad (4)$$

Demand Constraints

$$x_{11} + x_{21} + x_{31} = 12749.4$$

$$x_{12} + x_{22} + x_{32} = 6086.2$$

$$x_{13} + x_{23} + x_{33} = 10592.004$$

.

.

$$x_{91} + x_{92} + x_{93} = 3429100$$

Supply constraints

$$\begin{aligned} P_{11} + P_{12} + P_{13} + P_{14} + P_{15} + P_{16} + P_{17} + P_{18} + P_{19} + P_{110} \\ \leq 1,752,183 \end{aligned}$$

$$\begin{aligned} x_{21} + x_{22} + x_{23} + x_{24} + x_{25} + x_{26} + x_{27} + x_{28} + x_{29} + x_{210} \\ \leq 5,580,649 \end{aligned}$$

$$\begin{aligned} x_{31} + x_{32} + x_{33} + x_{34} + x_{35} + x_{36} + x_{37} + x_{38} + x_{29} + x_{310} \\ \leq 57,101,414 \end{aligned}$$

C_j the waste generation capacity of the all the collection radial was used in the demand constraint as shown in Table 1. K_i the capacity of the processing waste center that is to be opened was determined by selecting the minimum transportation cost function (d_{ij}) from Table 2 for the various 3 processing site. The values were summed across the row to determine the capacity each selected processing site.

3. Result of the study

3.1. Buffer map of livestock generation points

For the result of GIS transportation assessment of the study; Fig. 4 shows the buffering of the bio-waste sources in the study area for 5 km, 10 km, 15 km and 20 km distance. The result shows that most of the farms in the study area were situated within 5 km distance to each other.

The map shows suitable sites in majority of the study area.

3.2. Land use suitability map

The land use suitability map was obtained by weighted overlay

Table 2

Origin cost destination matrix (d_{ij}) for location allocation modelling in km.

d_{ij}	Onitsha	Orafite	Awka	Ihiala	Ajali	Akpo	Abagana	Adazi-ani	Aguleri	Omor
Onitsha	1	13.2327	34.9245	32.7025	48.1901	41.328	22.9573	24.1106	23.6949	45.8486
Orafite	13.2327	1	35.1095	34.2902	41.5605	31.318	23.9722	18.3885	33.264	55.0598
Awka	34.9245	35.1095	1	48.2437	23.3226	30.0606	12.0083	18.8496	25.2349	34.6648
<i>Ihiala</i>	<i>32.7025</i>	<i>34.2902</i>	<i>48.2437</i>	<i>1</i>	<i>44.7888</i>	<i>29.7022</i>	<i>39.6113</i>	<i>29.4073</i>	<i>52.6148</i>	<i>73.8375</i>
<i>Ajail</i>	<i>48.1901</i>	<i>41.5605</i>	<i>23.3226</i>	<i>44.7888</i>	<i>1</i>	<i>15.8605</i>	<i>29.0611</i>	<i>24.1861</i>	<i>47.195</i>	<i>57.6571</i>
Akpo	41.328	31.318	30.0606	29.7022	15.8605	1	29.2909	18.9998	48.2572	48.2572
<i>Abagana</i>	<i>22.9573</i>	<i>23.9722</i>	<i>12.0083</i>	<i>39.6113</i>	<i>29.0611</i>	<i>29.2909</i>	<i>1</i>	<i>11.7777</i>	<i>16.1172</i>	<i>19.085</i>
Adazi -ani	24.1106	18.3885	18.8496	29.4073	24.1861	18.9998	11.7777	1	11.9008	47.7151
Aguleri	23.6949	33.264	25.2349	52.6148	47.195	48.2572	16.1172	11.9008	1	22.1928
Omor	45.8486	55.0598	34.6648	73.8375	57.6571	48.2572	19.085	47.7151	22.1928	1

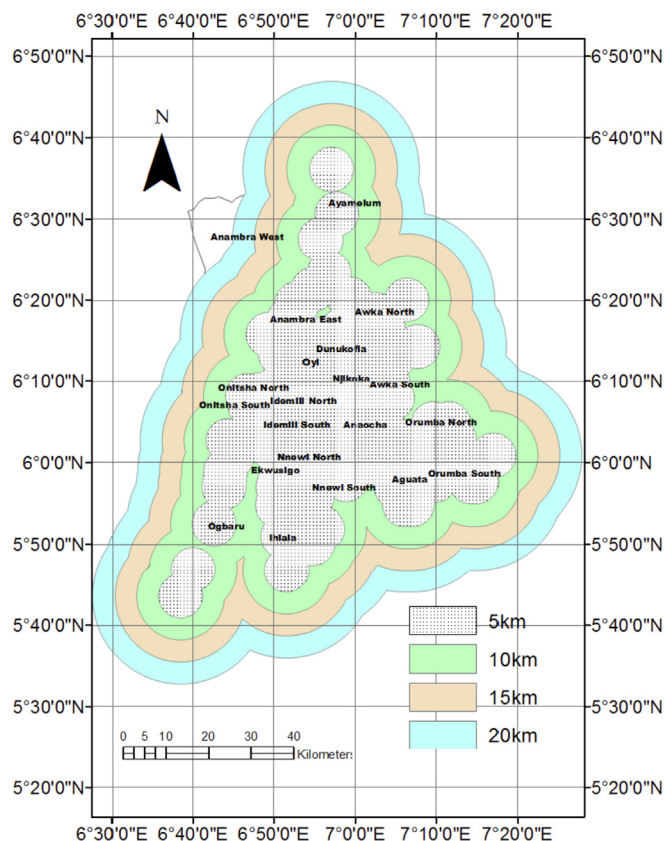


Fig. 4. Spatial buffering of livestock production sites.

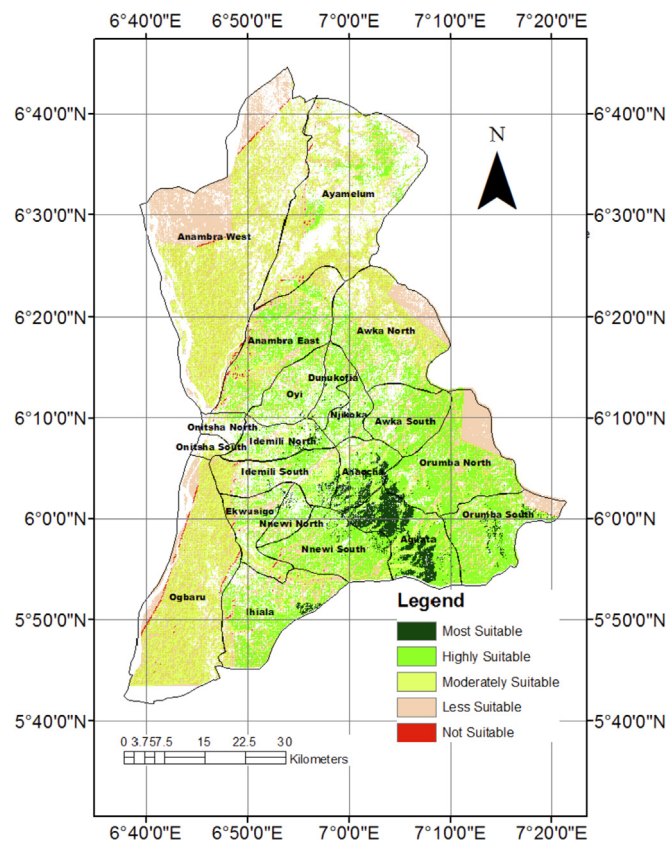


Fig. 5. Land use suitability map.

of necessary GIS data sets and excluding unwanted areas identified as constrain map; These are areas prone to flooding, areas that are close to residential areas and water bodies etc. The resultant dataset is the land use suitability map shown in Fig. 5. The Land use suitability analysis shown is graduated from the Most Suitable to the Less Suitable.

The suitability map suggests that the most suitable areas are located mainly in Aguata, Aniocha, Orumba South and North, Nnewi South and North L.G.A. Ogburu region is classified as moderately suitable area because of its low elevation which makes the area unsuitable for siting bio-energy plants considering the emergency of flood.

3.3. The minimum number of potential sites

The SCLM was used to determine the minimum number of processing sites needed to maximally process all the wastes in the study area using Microsoft Excel Solver. The result of the mathematical solution is as shown in Table 3.

The decision row above shows that a minimum of three locations are to be opened in Abagana, Ihiala and Ajali. A visualization of the locations in the map of the study area shows that the set covering location model provided a wide coverage for all the waste producing sites in the state. The three waste processing centers were all farther apart to ensure maximum coverage.

3.4. The location allocation of wastes to bioenergy centers

The location allocation model was further used to determine the allocation of the various wastes to the three multiple facility locations obtained using the set covering location as noted above. The

decision row above shows that a minimum of three locations are to be opened in Abagana, Ihiala and Ajali. The location allocation model as stated in equations (3) and (4) was solved with Excel Solver, the solution of the model is shown in Table 4.

Fig. 6 shows the location of the three facilities obtained using the integration of GIS, set cover and location/allocation models. A

Table 3

The result of sites selection using the set cover location model.

Objective Coefficient	13	6	11	2	2	1	13	1	4	Optimal objective value
	*1	*2	*3	*4	*5	*6	*7	*8	*9	
Decision Variable	0	0	0	1	1	0	1	0	0	17
Constraints	*1	*2	*3	*4	*5	*6	*7	*8	*9	
1	1	1	0	0	0	0	1	1	0	
2	1	1	0	1	0	0	0	0	0	
3	0	0	1	0	0	0	1	8	0	
4	0	1	0	1	0	0	0	0	0	
5	0	0	0	0	1	1	0	0	0	
6	0	0	0	0	1	1	0	1	9	
7	1	0	1	0	0	0	1	1	1	
8	0	0	0	0	0	1	1	1	0	
9	0	0	0	0	0	0	1	0	1	
Computations	*1	*2	*3	*4	*5	*6	*7	*8	*9	LHS RHS
1	0	0	0	0	0	0	1	0	0	1 1
2	0	0	0	1	0	0	0	0	0	1 1
3	0	0	0	0	0	0	1	0	0	1 1
4	0	0	0	1	0	0	0	0	0	1 1
5	0	0	0	0	1	0	0	0	0	1 1
6	0	0	0	0	1	0	0	0	0	1 1
7	0	0	0	0	0	0	1	0	0	1 1
8	0	0	0	0	0	0	1	0	0	1 1
9	0	0	0	0	0	0	1	0	0	1 1

Table 4
Location allocation model output.

P _{ij}	1 (Osha)	2 (Ora)	3 (Awka)	4 (Ihiala)	5 (Ajali)	6 (Akpo)	7 (Aba)	8 (Adazi)	9 (Agu)	10 (Omor)	Capacity
1 (Ihiala)	0	0	0	1752	0	0	0	0	0	0	1752
2 (Ajali)	0	0	0	0	1729	3851	0	0	0	0	5581
3 (Abagana)	12749	6086	10592	0	0	0	12976	11064	3429	205	57101
Waste q _{ity}	12749	6086	10592	1752	1729	3851	12976	11064	3429	205	

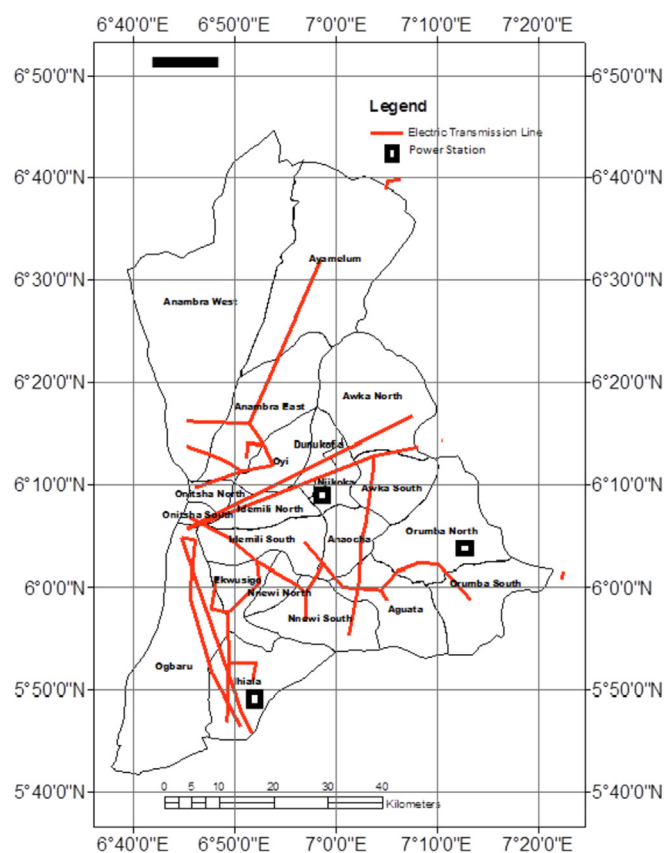


Fig. 6. Map of Suitable Power Plant Location and Transmission lines of the Study area.

visualization of the locations shows that the result of the study provided a wide coverage for all the waste producing sites in the state. The waste generation centers that seems to be uncovered are located in the Northern region of the study area, it is characterized with few and dispersive waste generation site. It is proposed that alternate waste utilization scheme should be set up for the region for economic purposes.

The output of the location model shows that a processing plant with about 5.6 tons processing capacity is best located at Ajali in Orumba North LGA (South Region); this will serve both Ajali and Akpo location radius. Then about 1.7 tons of waste generated within Ihiala location radius is best processed in a processing plant installed in Ihiala (East region). Finally a processing plant with 57 tons capacity is to be located at Abagana (central region), this will process waste generated from Oraifite, Awka, Adazi-ani, Aguleri and Omor location radial. This simply means that one big scale and two small scale biogas processing plant is to be installed in the study area to minimize the cost of transportation of agricultural wastes in the state.

4. Conclusion

The goal of this research work is to solve location allocation and site suitability problem by integrating geospatial technology and SCLM for optimized bioenergy locations considering economic advantages. GIS operations to obtain suitable locations were used to obtain land suitability map. Set cover was used to determine the minimum plants to cover the waste generation points, while location allocation model was used to allocate the waste generating point to waste processing centers. The result of the study using SCLM and location and allocation modelling indicates that a minimum of three locations are to be opened in the study area which should be sited in Abagana, Ihiala and Ajali. This implies that one big scale and two small scale biogas processing plant is to be installed in the study area to minimize the cost of allocation of agricultural waste in the state.

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